

AKTU Lucknow

REGULAR EXAMINATION ? 2025

Engineering Mathematics-II ? Regular Examination, 2025

Subject Code: MA201 Semester: 2 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Laplace Transforms)

1. Compare and contrast the key concepts involved in Laplace Transforms. [12 Marks]
2. Discuss the engineering applications of Laplace Transforms. [12 Marks]
3. Explain Laplace Transforms with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION B (Unit 2: Fourier Series)

4. List the advantages and limitations of Fourier Series. Explain any two in detail. [12 Marks]
5. Discuss the engineering applications of Fourier Series. [12 Marks]
6. Compare and contrast the key concepts involved in Fourier Series. [12 Marks]

SECTION C (Unit 3: Vector Integration)

7. How is Vector Integration implemented? Give an algorithm / procedure and explain. [12 Marks]
8. Define Vector Integration. Explain its significance in Engineering Mathematics-II with a suitable example. [12 Marks]
9. Define Vector Integration. Explain its significance in Engineering Mathematics-II with a suitable example. [12 Marks]

SECTION D (Unit 4: Complex Variables)

10. List the advantages and limitations of Complex Variables. Explain any two in detail. [12 Marks]
11. Compare and contrast the key concepts involved in Complex Variables. [12 Marks]
12. Solve a problem related to Complex Variables. Show all the steps clearly. [12 Marks]

SECTION E (Unit 5: Numerical Methods)

13. Compare and contrast the key concepts involved in Numerical Methods. [12 Marks]
14. Write short notes on Numerical Methods. [8 Marks]
15. List the advantages and limitations of Numerical Methods. Explain any two in detail. [12 Marks]

***** End of Question Paper *****